

-- Cox Regression - Sarcopenia Onset -----

i Definition : ewgsop2
i Covariates : time_dependent
i Dairy : continuous
i Route : mice
i Interaction: none

-- MICE route - pooling across imputed datasets --

Detected 20 imputed datasets.
i Dairy modelled as natural cubic spline (ns, df = 3).
Fitting .imp = 1 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1948 intervals | 49 events | 7801.6 person-years
Fitting .imp = 2 / 20 ...
Events: 57 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1948 intervals | 51 events | 7801.6 person-years
Fitting .imp = 3 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1949 intervals | 52 events | 7804.1 person-years
Fitting .imp = 6 / 20 ...
Events: 57 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1948 intervals | 51 events | 7801.6 person-years
Fitting .imp = 7 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1948 intervals | 50 events | 7801.6 person-years
Fitting .imp = 8 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1948 intervals | 49 events | 7801.6 person-years
Fitting .imp = 9 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1948 intervals | 50 events | 7801.6 person-years
Fitting .imp = 10 / 20 ...
Events: 58 sarcopenia cases out of 1121 participants

??? Time-dependent dataset: 1948 intervals | 50 events | 7801.6 person-years
Fitting .imp = 12 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1948 intervals | 50 events | 7801.6 person-years
Fitting .imp = 13 / 20 ...
Events: 57 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1948 intervals | 51 events | 7801.6 person-years
Fitting .imp = 14 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1948 intervals | 50 events | 7801.6 person-years
Fitting .imp = 15 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

??? Time-dependent dataset: 1948 intervals | 50 events | 7801.6 person-years
Fitting .imp = 16 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

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??? Time-dependent dataset: 1948 intervals | 51 events | 7801.6 person-years
Fitting .imp = 18 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants
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-- Time-dependent covariates dataset (one row per pt x visit) --
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??? Time-dependent dataset: 1948 intervals | 49 events | 7801.6 person-years
Fitting .imp = 19 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants
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-- Time-dependent covariates dataset (one row per pt x visit) --
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```
??? Time-dependent dataset: 1948 intervals | 49 events | 7801.6 person-years
Fitting .imp = 20 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants
```

```
-- Time-dependent covariates dataset (one row per pt x visit) --
```

```
??? Time-dependent dataset: 1948 intervals | 50 events | 7801.6 person-years
```

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-- Pooled adjusted model results (Rubin's rules) --
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```
# A tibble: 10 x 5 term HR CI_low CI_high p.value <chr> <dbl> <dbl> <dbl> <dbl>
1 splines::ns(dairy_exposure, df = 3)1 5.25 0.426 64.8 0.189 2
splines::ns(dairy_exposure, df = 3)2 0.0743 0.000208 26.5 0.376 3
splines::ns(dairy_exposure, df = 3)3 0.00546 0.000000218 137. 0.304 4 BMI
0.710 0.633 0.795 0.000000459 5 education_levelMedium (ISCED 3-4) 0.878 0.380
2.03 0.754 6 education_levelHigh (ISCED 5-8) 0.900 0.290 2.80 0.852 7
smoking_statusFormer 0.895 0.477 1.68 0.722 8 smoking_statusCurrent 0.796 0.288
2 20 0 652 9 myna min day f1 0 999 0 996 1 00 0 496 10 diabetes statusDiabetes
```

-- Assumption checks [mice_imp1] --

-- 1. Proportional Hazards - cox.zph() | pass: global p > 0.05

chisq df p splines::ns(dairy_exposure, df = 3) 4.6505 3 0.20 BMI 0.0524 1 0.82
education_level 3.2629 2 0.20 smoking_status 3.7984 2 0.15 mvpa_min_day_f1
0.2468 1 0.62 diabetes_status 2.0317 1 0.15 GLOBAL 15.3415 10 0.12
??? PH assumption not violated (global p = 0.1201)
Schoenfeld plots saved to
03_outputs/Cox/mice_ewgsop2_time_dependent_continuous_spline3_27072026_1255

-- 2. Linearity - martingale (null model) vs continuous predictors | pass: smooth ??? linear

'geom_smooth()' using formula = 'y ~ x'
'geom_smooth()' using formula = 'y ~ x'
Linearity plots saved to
03_outputs/Cox/mice_ewgsop2_time_dependent_continuous_spline3_27072026_1255

-- 3. Outliers - deviance & martingale residuals | pass: |dev| <= 3

i 1 observation(s) with |deviance residual| > 3.

-- 4. Influential observations - std. DFBetas | pass: no extreme spikes

i 25 obs exceed |DFBeta_std| > 0.2.

-- 5. Overall Fit - Cox-Snell residuals | pass: points on 45° diagonal

Cox-Snell plot saved to
03_outputs/Cox/mice_ewgsop2_time_dependent_continuous_spline3_27072026_1255

-- 6. Multicollinearity - VIF | pass: VIF < 5 (< 10 relaxed)

GVIF Df GVIF^(1/(2*Df)) splines::ns(dairy_exposure, df = 3) 1.110 3 1.018 BMI
1.126 1 1.061 education_level 1.160 2 1.038 smoking_status 1.057 2 1.014

-- Adjusted scenario survival curves (time-dependent) [mice_imp1]

??? Spline HR plot saved:

03_outputs/Cox/mice_ewgsop2_time_dependent_continuous_spline3_27072026_1255/mice_spline_hr.png

-- Linearity test: spline vs linear dairy term

i $\chi^2(2) = 1.51$, $p = 0.469$

No significant departure from linearity ($p \geq 0.05$).

-- Incidence rates by dairy quartile [mice]

quartile n_pts n_events person_years rate_per_1000py ci_low ci_high 1 Q1 (low)

281 10.00 1965.30 5.09 2.44 9.36 2 Q2 280 9.70 1967.92 4.93 2.33 9.15 3 Q3 280

13.00 1934.70 6.72 3.58 11.49 4 Q4 (high) 280 17.45 1933.80 9.02 5.30 14.36

Incidence table saved to

03_outputs/Cox/mice_ewgsop2_time_dependent_continuous_spline3_27072026_1255

Model results saved to

03_outputs/Cox/mice_ewgsop2_time_dependent_continuous_spline3_27072026_1255/mice_model_results.csv